

Mandatory Assignment 1

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Efficient portfolios and estimation uncertainty

Exercise 1.1

We download daily price data for the S&P 500 (`^GSPC`) and the 13-week T-bill yield (`^IRX`) from Yahoo Finance over January 2000 to February 2026, and then convert both series to month-end observations. Monthly market returns are computed from the S&P 500 price series, while the monthly risk-free rate is obtained by converting the annualized T-bill yield via $r_{f,t} = (1 + \text{IRX}_t/100)^{1/12} - 1$. The monthly excess market return is then constructed as $\tilde{r}_{m,t} = r_{m,t} - r_{f,t}$. Table 1 reports the estimated monthly and annualized moments of $\tilde{r}_{m,t}$.

Table 1: Monthly and annualized mean excess return and variance of the S&P 500

	monthly_mean	monthly_var	annualized_mean	annualized_var
S&P500	0.0045	0.0019	0.0545	0.0228

Exercise 1.4

We implemented a reusable mean–variance routine that, given a mean vector μ and covariance matrix Σ , computes the mean–variance efficient frontier (the set of portfolios with minimum variance for each target expected excess return). In the same function we also compute the tangency portfolio, i.e. the portfolio that maximizes the Sharpe ratio when working with excess returns, with weights proportional to $\Sigma^{-1}\mu$ and normalized to sum to one.

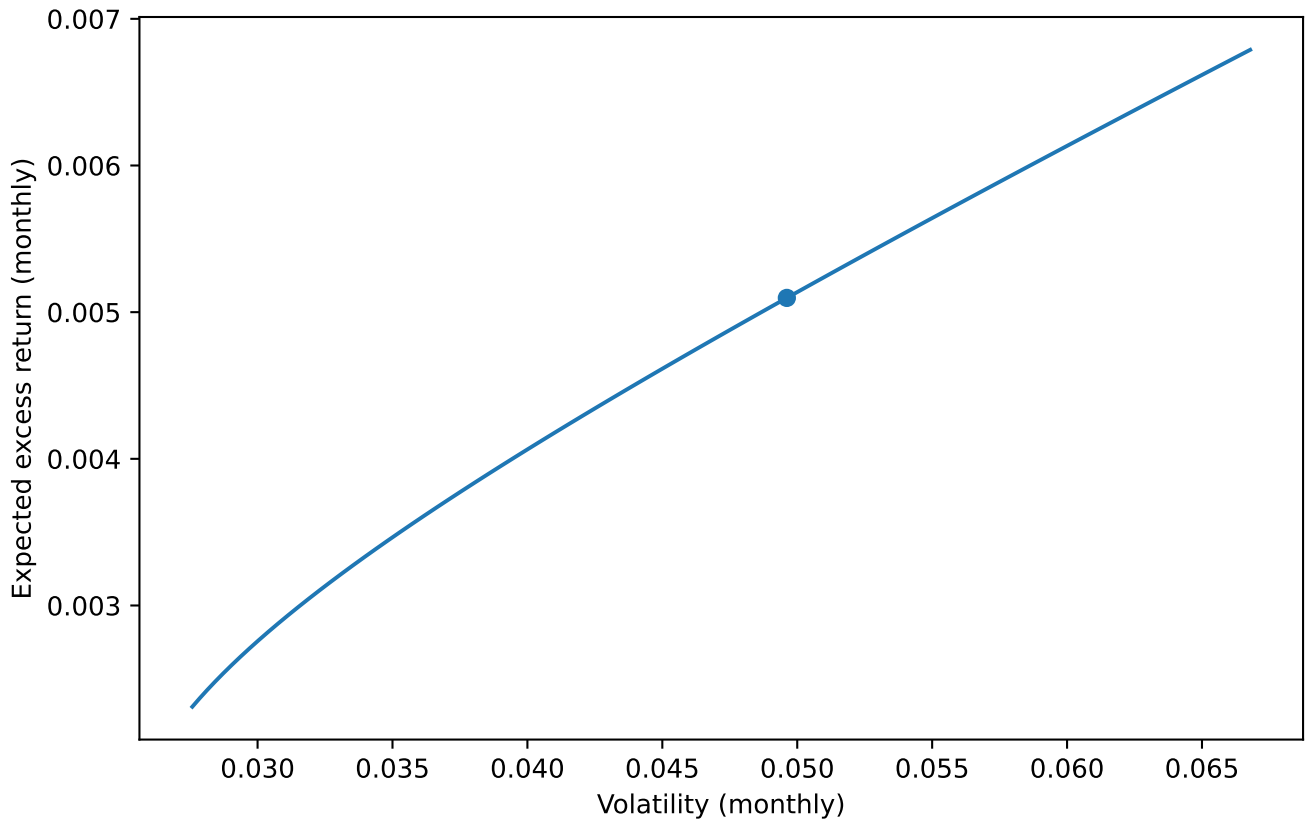


Figure 1: Efficient Frontier (True μ , Σ) + Tangency Portfolio

Using the “true” parameters (μ, Σ) from Exercise 2, we plotted the corresponding efficient frontier in mean–volatility space in Figure 1. The curve shows the best attainable trade-off between expected monthly excess return (y-axis) and monthly volatility (x-axis). The marked point is the tangency portfolio, which lies on the frontier and delivers the highest risk-adjusted performance (maximum Sharpe) among all fully invested portfolios in this setting.

Exercise 1.5

In this exercise we study how sampling uncertainty in $(\hat{\mu}, \hat{\Sigma})$ affects the estimated mean–variance frontier. Using simulated samples of length T drawn from the true distribution $\mathcal{N}(\mu, \Sigma)$, we re-estimate $(\hat{\mu}, \hat{\Sigma})$ and recompute the efficient frontier. The resulting estimated frontier can differ substantially from the true frontier when T is small, because the optimizer exploits estimation noise in the sample moments. This creates an “optimism” effect: the estimated frontier may appear too attractive in-sample and the implied portfolio weights are unstable, see Figure 2.

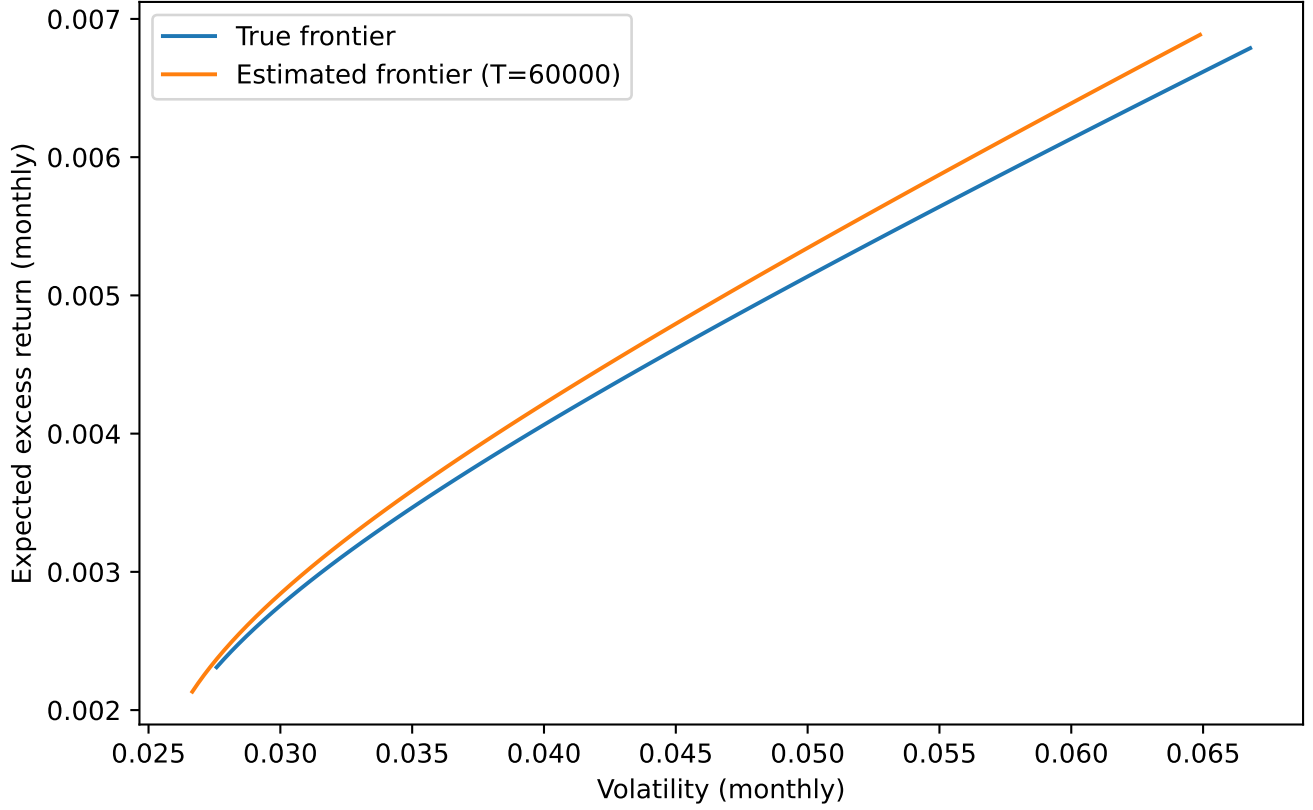


Figure 2: Efficient Frontier True vs Estimated

The Monte Carlo plots in Figure 3 illustrate that estimation error decreases rapidly as T grows. When T is large relative to the number of assets N , the cloud of estimated frontiers collapses toward the true frontier, and for very large T the two frontiers are nearly indistinguishable. Overall, the severity of estimation error is governed primarily by the ratio N/T (and, in practice, also by how ill-conditioned Σ is): higher T or lower N makes the estimated frontier much more reliable.

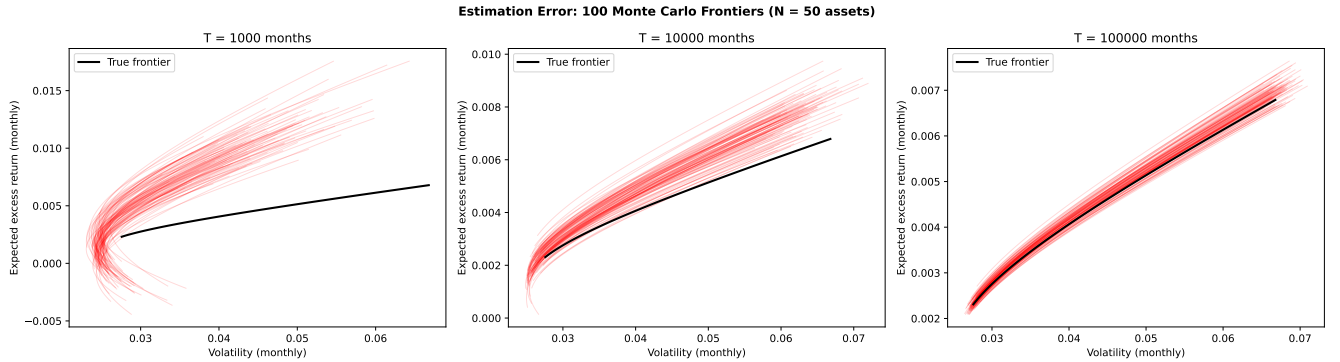


Figure 3: Three different Estimation Error Monte Carlo plots

Exercise 1.6

In this exercise we evaluate the out-of-sample performance of the plug-in tangency portfolio. For each sample length $T \in \{60, 120, 240, 600, 1200\}$ we simulate $S = 100$ independent samples from the true return distribution $\mathcal{N}(\mu, \Sigma)$. In each sample we estimate $(\hat{\mu}, \hat{\Sigma})$, compute the plug-in tangency weights $\hat{w} \propto \hat{\Sigma}^{-1} \hat{\mu}$, and then evaluate performance using the *true* parameters, i.e. $SR(\hat{w}) = \frac{\hat{w}^\top \mu}{\sqrt{\hat{w}^\top \Sigma \hat{w}}}$.

Figure 4 and Table 2 show that the plug-in tangency portfolio performs poorly when T is small: average out-of-sample Sharpe ratios are close to zero and the dispersion across simulations is large. As T increases, the mean out-of-sample Sharpe rises and the uncertainty shrinks, but even at moderate sample sizes the plug-in approach remains noticeably below the true tangency Sharpe. This pattern reflects the sensitivity of mean–variance optimization to estimation error, especially in $\hat{\mu}$ and in the inversion of $\hat{\Sigma}$.

Overall, performance is primarily governed by the effective signal-to-noise ratio in the estimation problem. When T is small relative to N , many covariance parameters must be inferred from limited data, which produces unstable portfolio weights and unreliable out-of-sample outcomes. Increasing T (or reducing N) improves stability and brings the plug-in portfolio closer to the true optimum.

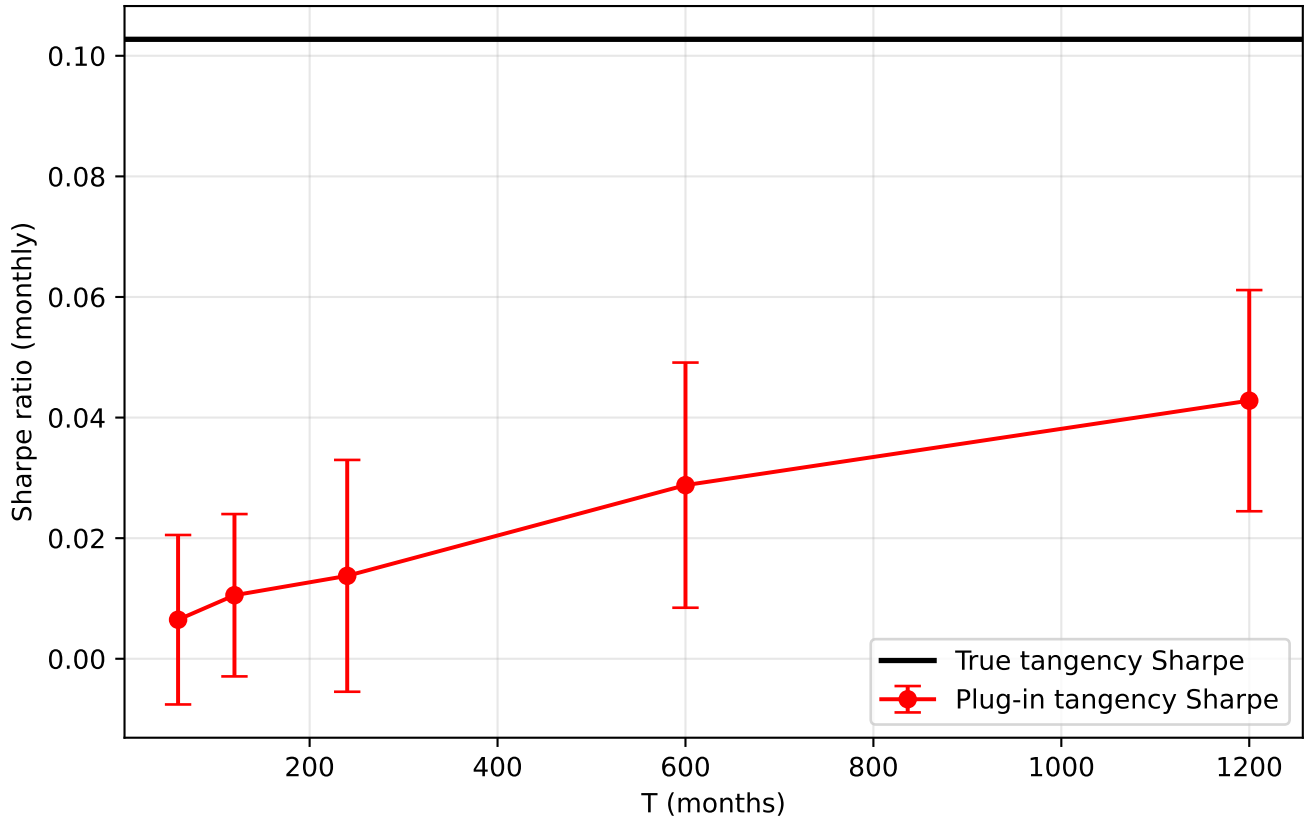


Figure 4: Out-of-Sample Sharpe

Table 2: Average out-of-sample Sharpe ratio and standard deviation of the plug-in tangency portfolio across sample lengths T .

	T	mean_sharpe	std_sharpe
0	60	0.006475	0.014053
1	120	0.010536	0.013462
2	240	0.013753	0.019223
3	600	0.028793	0.020330
4	1200	0.042805	0.018341

Exercise 1.7

In addition to the plug-in tangency portfolio, we consider three simple alternatives that are designed to be more robust to estimation error.

Equal-weight (1/N). We set $w_i^{EW} = 1/N$ for all assets. This strategy requires no estimation and therefore provides a stable diversification baseline.

Minimum variance (GMV). We compute the global minimum-variance portfolio $w^{GMV} \propto \hat{\Sigma}^{-1} \mathbf{1}$ with $\mathbf{1}^\top w^{GMV} = 1$. Because GMV relies only on the covariance matrix and does not use expected returns, it is typically less sensitive to the severe noise in $\hat{\mu}$.

Inverse volatility (IVOL). We set weights proportional to inverse estimated volatility, $w_i^{IVOL} \propto 1/\hat{\sigma}_i$ where $\hat{\sigma}_i = \sqrt{\hat{\Sigma}_{ii}}$ and $\mathbf{1}^\top w^{IVOL} = 1$. This rule uses only diagonal risk estimates and avoids estimating correlations and expected returns, which makes it particularly stable.

We embed all strategies in the simulation framework from Exercise 5. For each sample length T , we estimate the required inputs from simulated data and then evaluate out-of-sample performance using the true parameters (μ, Σ) . Figure 5 and the accompanying table show a clear ranking by robustness: the plug-in tangency portfolio improves as T increases, but it remains the most sensitive to estimation error. In contrast, equal-weight and inverse-volatility are highly stable across T , while minimum-variance lies in between—more robust than the plug-in tangency portfolio, but still affected by covariance estimation noise.

It is worth mentioning, that when the available sample size is limited relative to the number of assets, sophisticated mean–variance optimization tends to amplify estimation noise. In such settings, simple diversification and risk-based allocations can deliver more reliable out-of-sample performance than plug-in mean–variance portfolios.

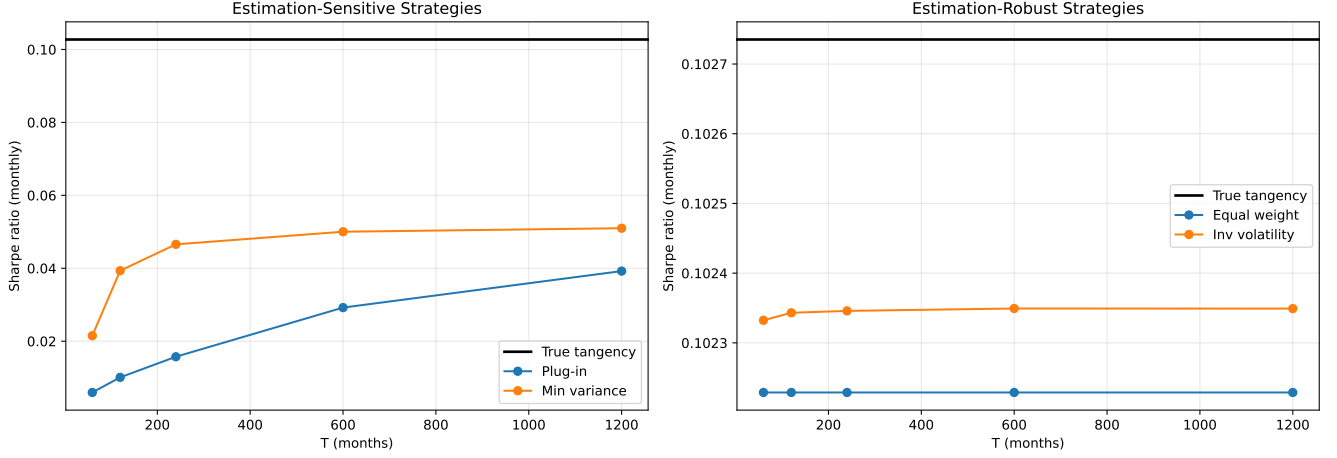


Figure 5: Out-of-Sample Strategy Comparison

Table 3 complements Figure 5 by reporting the average out-of-sample Sharpe ratio for each strategy at each sample length T . The plug-in tangency portfolio improves monotonically with T , consistent with decreasing estimation error, but it remains well below the true tangency benchmark even at $T = 1200$. In contrast, equal-weight and inverse-volatility deliver stable Sharpe ratios across all T , indicating high robustness to estimation noise. The minimum-variance strategy improves substantially relative to the plug-in approach, but still depends on covariance estimation and therefore benefits from larger T .

Table 3: Average out-of-sample Sharpe ratio by strategy across sample lengths

	T (months)	Plug-in	Equal Weight	Min Variance	Inv Volatility
0	60	0.0059	0.1022	0.0215	0.1023
1	120	0.0101	0.1022	0.0393	0.1023
2	240	0.0157	0.1022	0.0466	0.1023
3	600	0.0292	0.1022	0.0500	0.1023
4	1200	0.0392	0.1022	0.0510	0.1023

Conclusion

Using historical S&P 500 excess returns and a single-factor market model, we constructed a synthetic universe with “true” parameters (μ, Σ) and used it as a controlled laboratory for portfolio choice. We showed that mean–variance objects (the efficient frontier and the tangency portfolio) are highly sensitive to estimation error: when the sample size T is small relative to the number of assets N , the estimated frontier varies widely and plug-in tangency weights overfit noise, leading to weak and unreliable out-of-sample Sharpe ratios when evaluated under the true parameters.

Across simulations, performance improves as T increases, but simple rules and risk-focused allocations are more robust to limited data. In particular, equal-weighting, inverse-volatility weighting, and minimum-variance portfolios deliver more stable out-of-sample results than the plug-in tangency portfolio in small-to-moderate samples. The practical takeaway is that with realistic sample sizes,

reducing estimation risk (via simplicity or shrinkage/risk-based rules) is often more valuable than aggressively optimizing noisy inputs.